

Computer Hacking Forensic Investigator (CHFI)

Version 11

Lab Report

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Course Name: Computer Hacking Forensic Investigator (CHFI V11)

Report Title: CHFI Lab Report (Module 14)

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Lab Scenario

Catherine, an incident handling manager at a brokerage firm, was notified of shares purchased at considerably high prices by the firm. As an incident manager, she suspected a cybercrime and reported this to the FBI. Catherine mentioned that the firm's accounts were also suspended due to non-payment towards the shares purchase.

A team of cybercrime investigation experts visited the firm and started their initial investigation. Later, the team found that unknown individuals had hacked mobile devices that were used to purchase the shares at higher prices using software from the firm's network. The experts identified one individual by the mobile device address used in the purchase as well as the service used.

Lab Objectives

The objective of this lab is to offer comprehensive information on mobile forensics. The detailed objectives of this lab include the following:

- Examining an Android image file and retrieving carved data from it

Overview of Mobile Forensics

The need for mobile forensics has increased due to the prevalence of mobile devices, resulting in significant incidences of mobile device crimes and the related electronic evidence stored on these devices. The primary objectives of mobile forensics are to identify, extract, document, and interpret electronic data. It is always recommended to outline and define policies and procedures to be followed

to carry out mobile forensics. Data must be retrieved and analyzed without causing any damage and ensuring its authenticity.

Lab Tasks

Recommended lab to assist you in mobile forensics:

- Analyze Android Forensic Image and Carve Deleted Files

Lab 1: Analyze Android Forensic Image and Carve Deleted Files

Lab Scenario

The cybercrime police arrested Joe Smith, who was involved in nefarious activities, and seized his mobile device. To charge the accused with a crime, the police have to submit evidence to proceed with the case. Therefore, they appointed a digital forensic investigator to examine the mobile device for evidence. In this case, recovering deleted data is crucial as it might contain evidence required for the case. To carve deleted contents from the mobile device, the forensic investigator acquires dd image of the device contents and examines the image file for the required evidence.

Being a mobile forensics expert, you need to have thorough knowledge of carving data from hex codes and various file types.

Lab Objectives

File carving is the process of retrieving deleted information from a hard disk or an image in the absence of the filesystem that originally created the file.

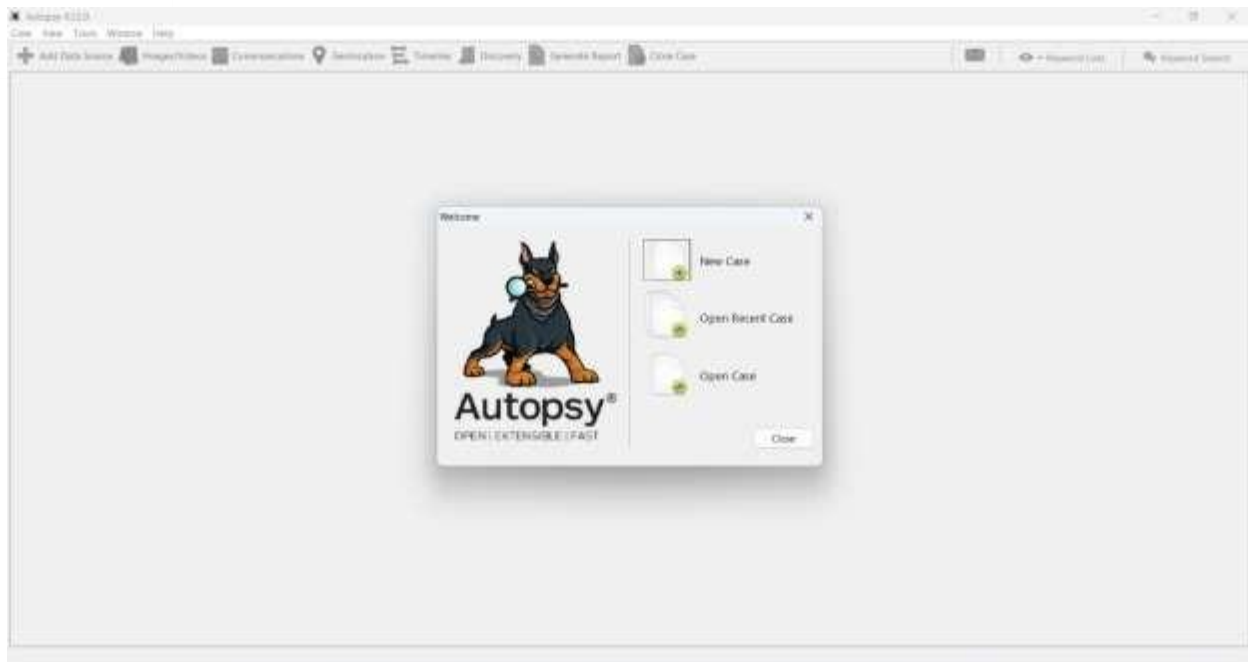
In this lab, you will learn how to examine an image file obtained from a mobile device and retrieve deleted data from it.

Overview of the Lab

This lab familiarizes you with the process of examining an evidence file obtained from an Android mobile device and retrieving the carved files from it.

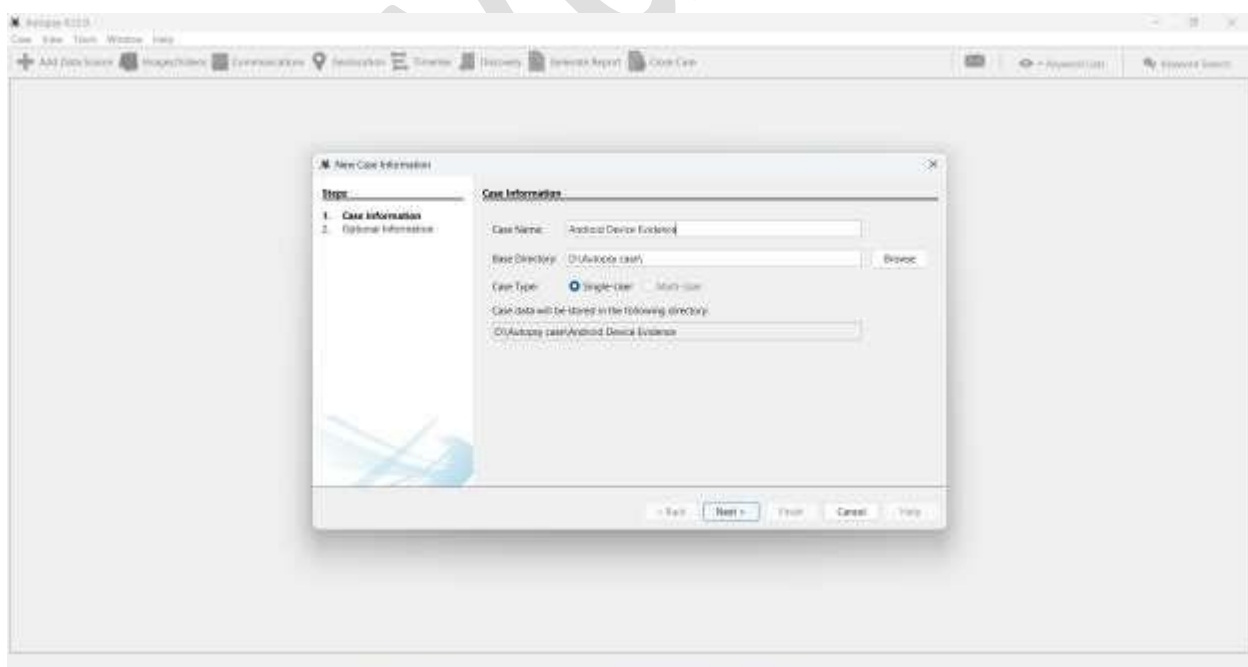
Lab Tasks

1. Launch the Autopsy 4.22.0 . Click on create a new case.

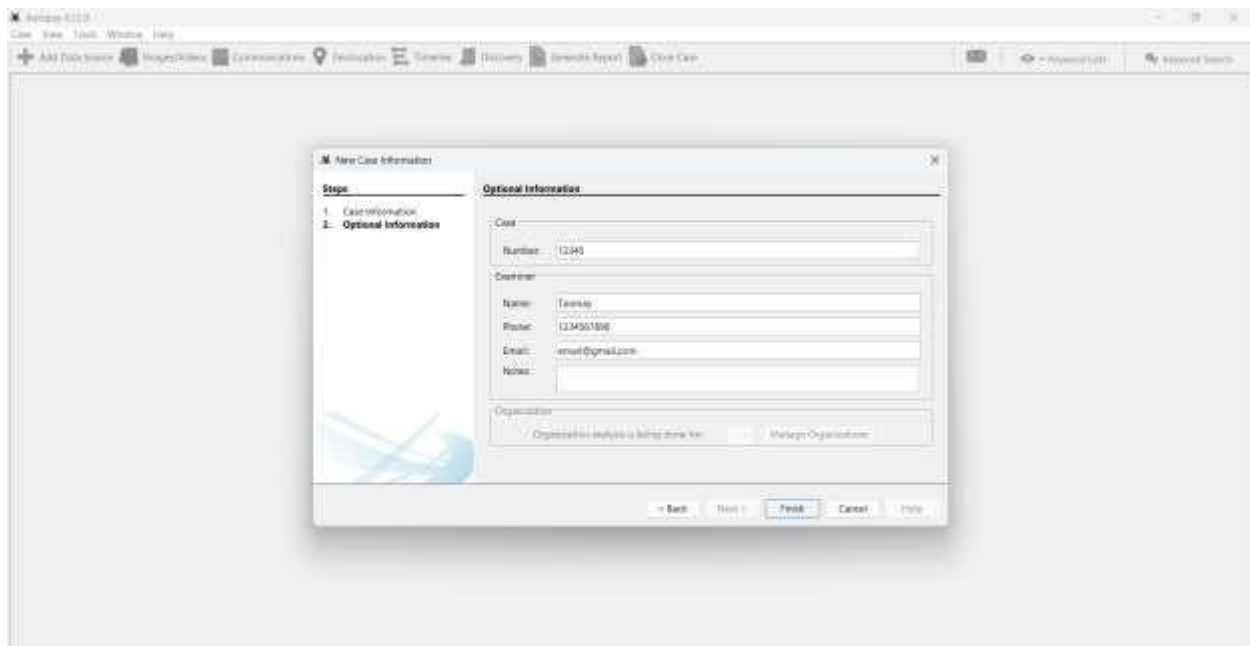


2. The New Case Information window will appear; enter a case name in the Case Name text field and provide a path for the Base Directory. The base directory is the location where the case data will be stored. The case name may be entered according to your identification purpose. In this lab, we are assigning the case name as Android Device Evidence.

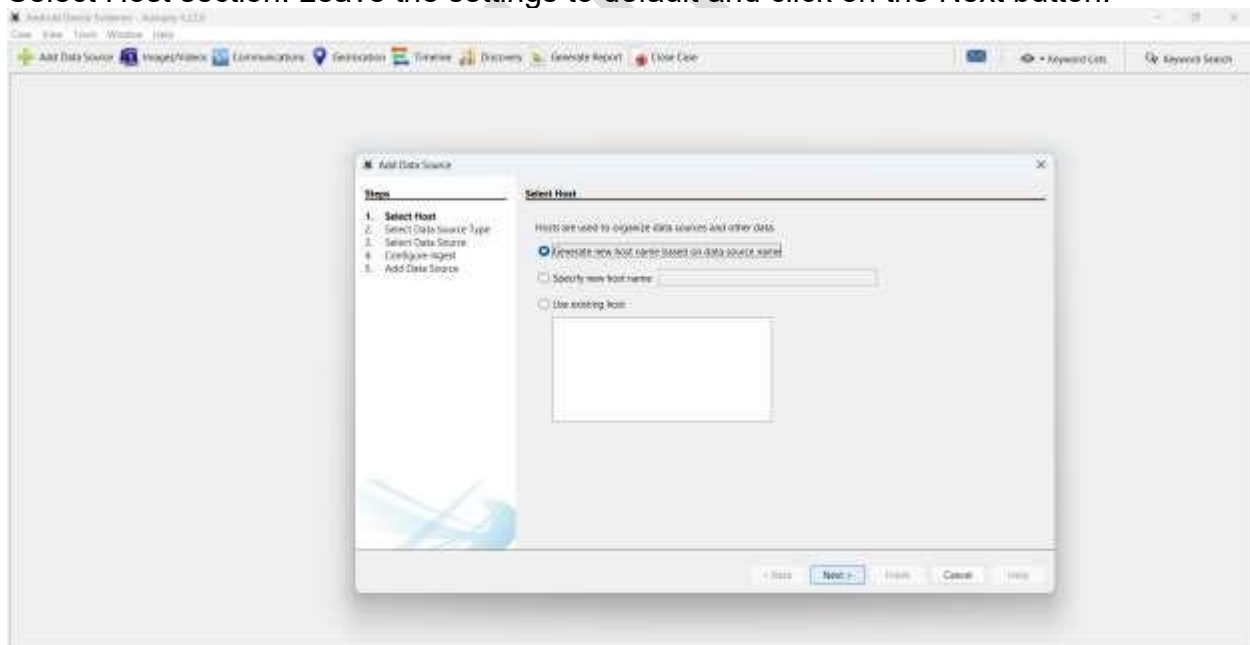
3. Here, we are saving the case information in the Autopsy folder located on the D Drive. Therefore, assign this folder as the Base Directory and click the Next button.



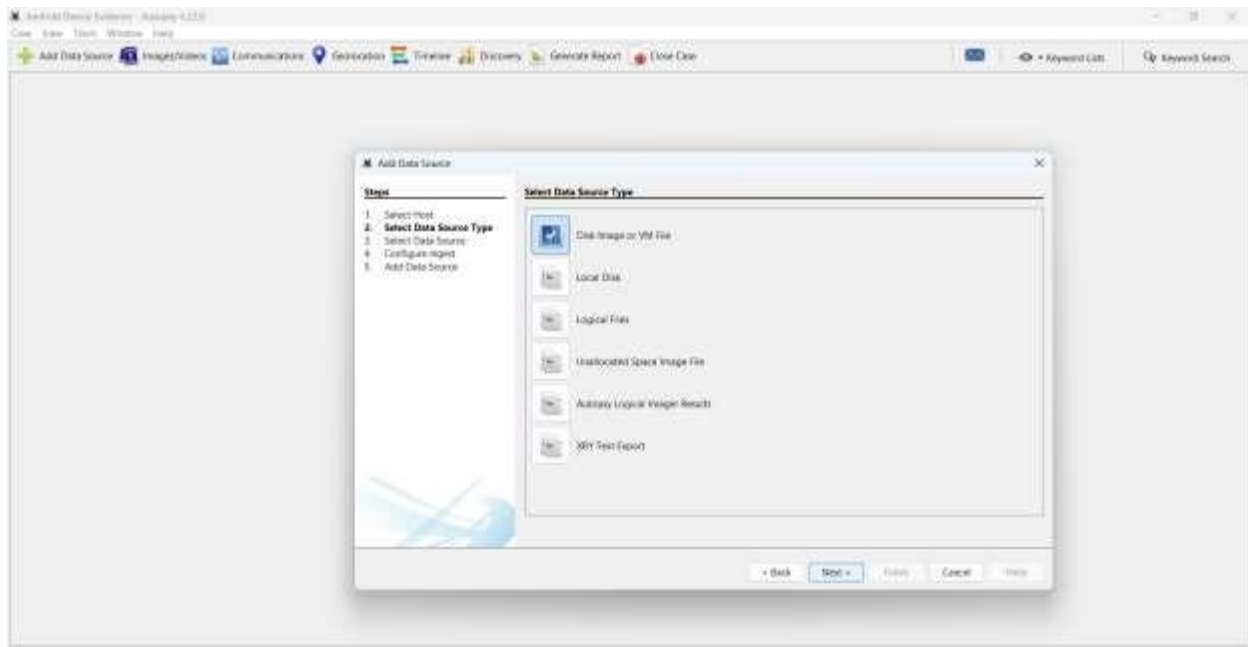
4. The Optional Information section will appear; provide the Case Number and Examiner details (here, we have provided Case Number as 12345). Enter the details of the Examiner and click the Finish button.



5. Once you click the Finish button, the application will take some time to load the data after which the Add Data Source window will appear. In this window, you will first see the Select Host section. Leave the settings to default and click on the Next button.

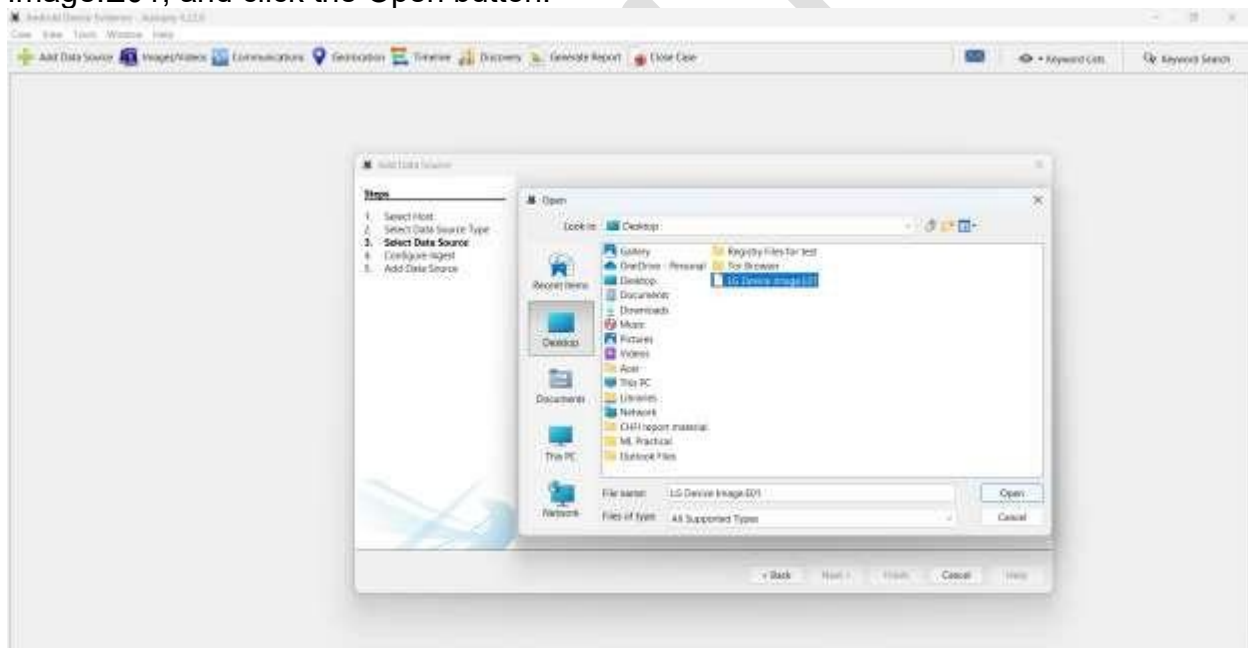


6. Next in the Select Data Source Type section, ensure that the Disk Image or VM File option is selected and click the Next button.

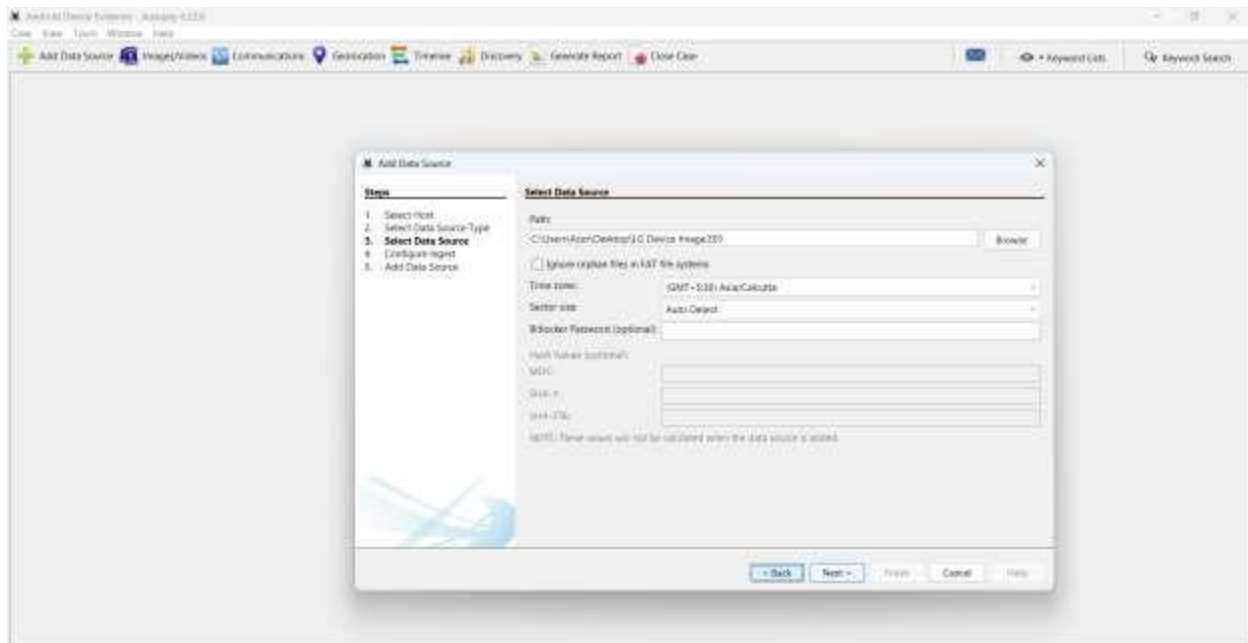


7. The Select Data Source section will appear. Click Browse.

8. An Open window will appear. Navigate to the location Desktop, select the file LG Device Image.E01, and click the Open button.

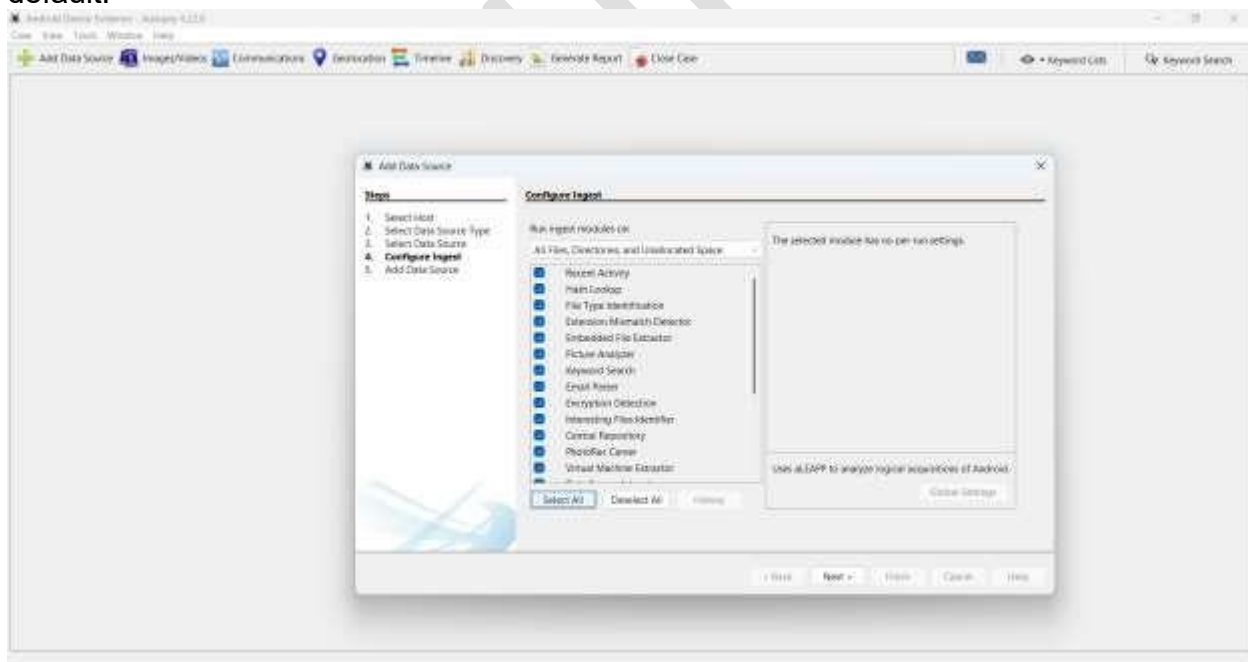


9. In the Select Data Source section, the directory of the selected evidence file (LG Device Image.E01) will now be displayed. Click the Next button.

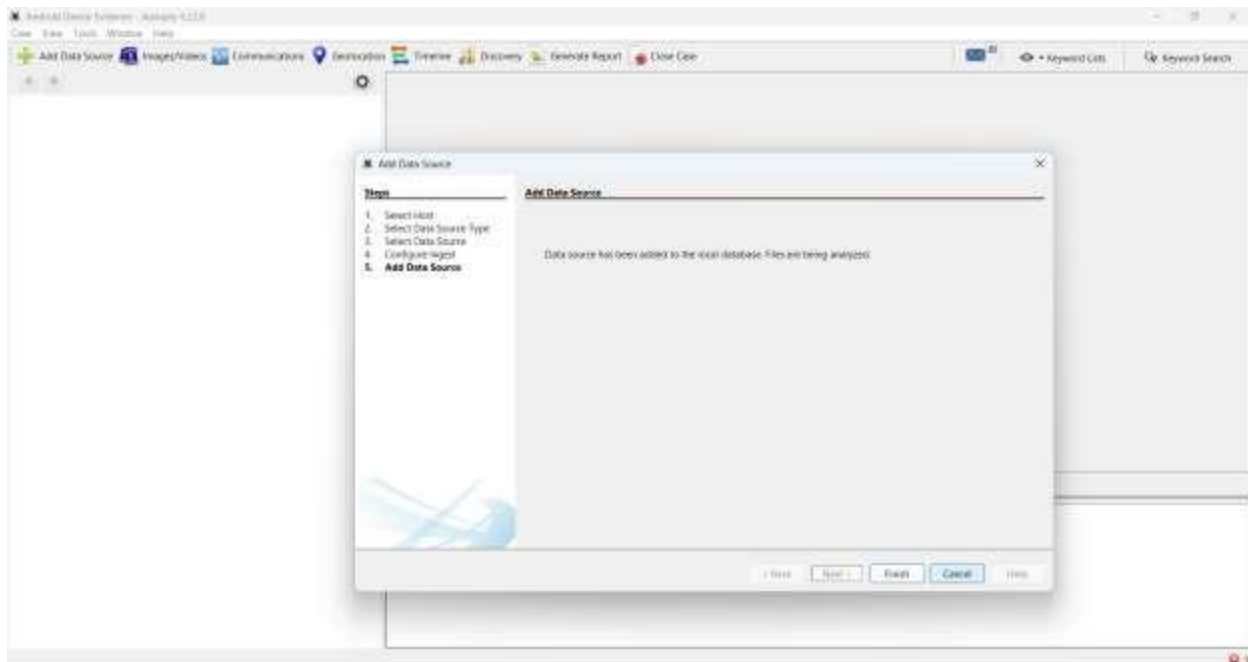


10. The Configure Ingest section will now appear where you will see a list of modules that are checked in the Run Ingest Modules on: column. Leave these modules checked by default and click the Next button.

When you select a module, it displays options specific to the selected module in the right pane. Here, you can configure settings for each module. In this lab, we set the options to default.

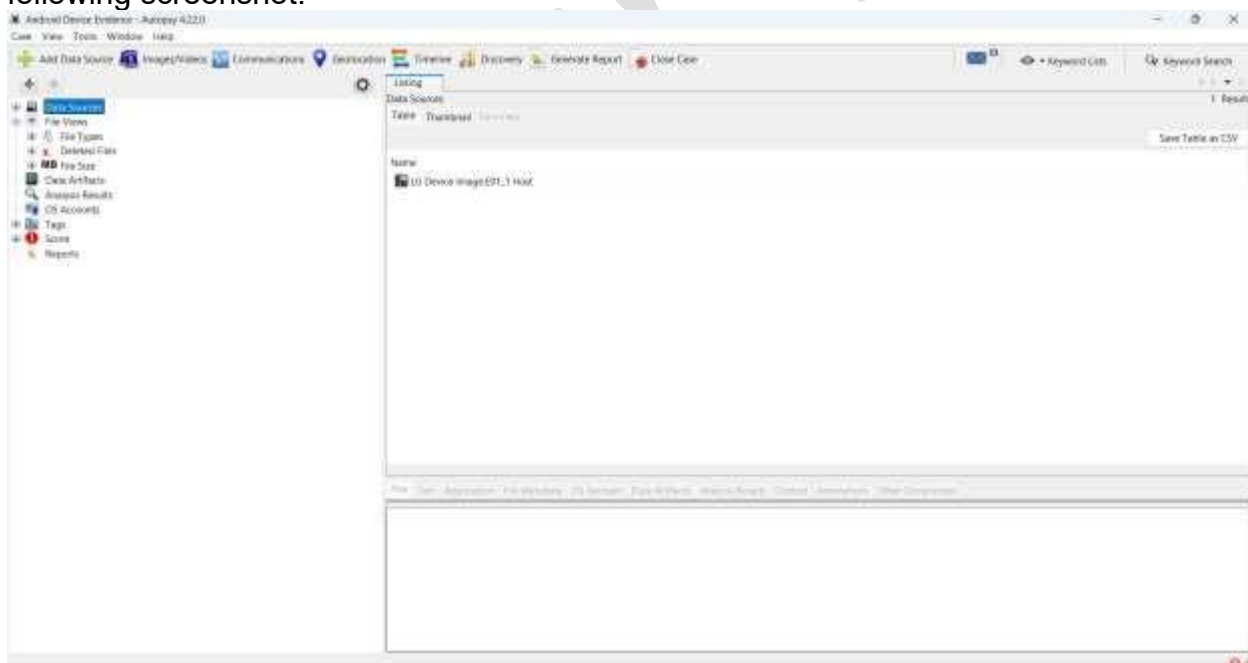


11. The Add Data Source section will appear displaying a message that reads Data source has been added to the local database. Files are being analyzed. Click the Finish button.

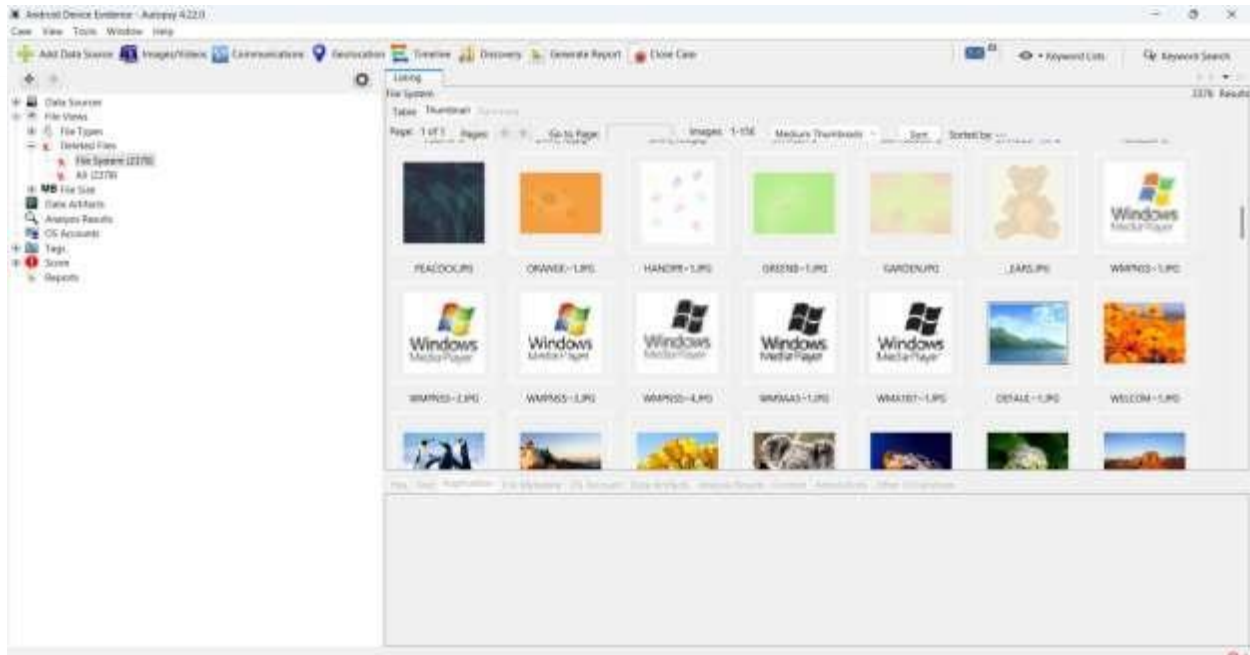


Autopsy takes time to analyze the image file. View the analysis status at the lower right corner of the Autopsy window. Once the analysis is complete, the status disappears.

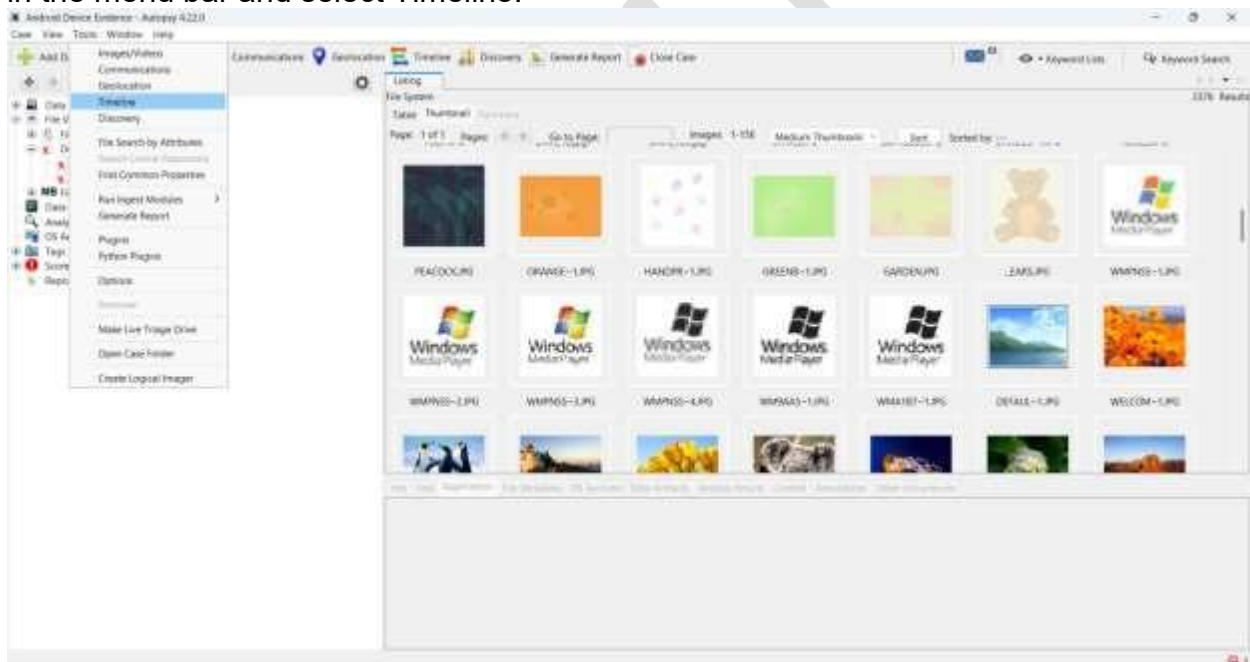
12. After analyzing all the files, Autopsy displays the retrieved information as shown in the following screenshot:



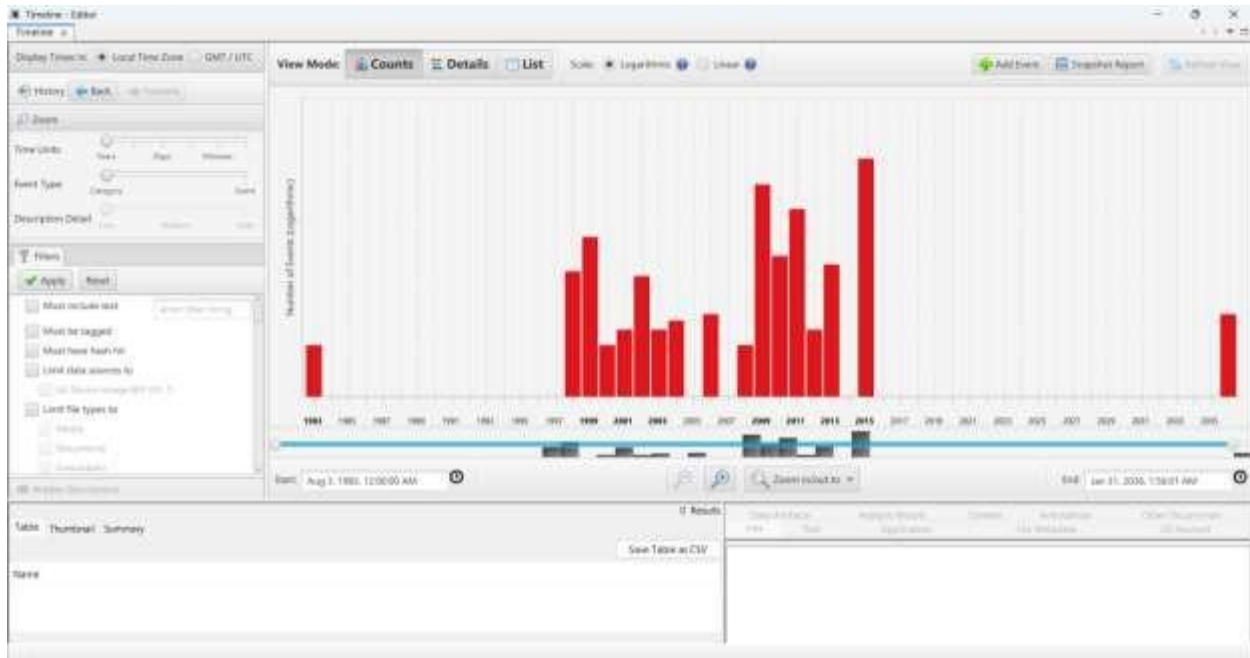
13. Navigate to file systems. Here you will see all the data in the device with their hex properties, Text properties, etc.



14. The Timeline feature of Autopsy helps you retrieve information on various events that occurred on the device during a selected time period. To view the timeline, select Tools in the menu bar and select Timeline.



15. A timeline window will appear, displaying a logarithmic graph that contains information on various events that occurred during a given time interval.



16. In this window, you can specify a particular time interval to retrieve information on the events or activities that occurred within that specified time interval.

17. The red colored bars in the above graph represent file system activity.

18. In this manner, you can analyze an image and examine the file system of the target device during the process of a forensic investigation.

19. Close all open windows.